Short Notes: Biomolecules of Life - Proteins

Definition: Proteins are the most abundant organic molecules of life. The term originates from the Greek word "Proteios", meaning "holding the first place".

They are heteropolymers made up of linear chains of amino acids (generally >50 to several hundred units).

1. Elemental Composition of Proteins

- Carbon (C): 50 - 55%
- Hydrogen (H): 6 - 7.3%
- Oxygen (O): 19 - 24%
- Nitrogen (N): 13 - 19%
- Sulphur (S): 0 - 4%
- Trace elements: P, Fe, Cu, I, Mg, Mn, Zn

2. Biological Functions

Proteins fold into diverse 3-D shapes (spherical, ellipsoidal, long strands) to perform specific jobs. Shape matters!

- Support / Structural: Collagen (skin), Keratin (hair, nails).
- Defense: Antibodies, venom.
- Transport: Hemoglobin (transports oxygen in blood).
- Motion: Actin & Myosin (muscle contraction).
- Regulation / Hormones: Insulin (controls blood sugar).
- Enzyme Catalysis: Pepsin (digestion), Hexokinase (glycolysis).
- Other roles: Acid-base balance, fluid balance, blood clotting.

3. Structural Organization

All information for 3-D structure is contained in the linear sequence of amino acids.

- Primary (**1°**): Linear amino acid sequence.
- Secondary (**2°**): Regular sub-structures (e.g., **α**-helices, **β**-sheets).
- Tertiary (**3°**): Three-dimensional folded structure.
- Quaternary (**4°**): Complex of multiple protein molecules interacting.

4. Classification of Proteins

- **Simple Proteins:** Made ONLY of amino acids. Yield only amino acids upon hydrolysis.
  - Fibrous: Ligament, cartilage, hair, nail.
  - Globular: Albumin, Globulin, Histone, Gliadin.
- **Conjugated Proteins:** Simple proteins combined with a non-protein substance (called a prosthetic group or cofactor).
  - Nucleoprotein (RNA/DNA), Lipoprotein (Lipid), Glycoprotein (Carbohydrate), Hemoglobin (Heme/Iron).
- **Derived Proteins:** Not naturally occurring. Obtained from simple proteins by action of enzymes/chemicals (e.g., Peptones, Peptides).

5. Amino Acids: The Building Blocks

There are 20 commonly found amino acids that make up living organisms. They share a common central structure.

General Structure

A central "Alpha carbon" (C_α) attached to 4 groups:

1. An Amino group (**-NH_2**)
2. A Carboxyl group (**-COOH**)
3. A Hydrogen atom (**-H**)
4. A Variable *R-group* (Side chain) - This makes each amino acid unique!

Chirality & Optical Isomerism

The C_α has four different substituents, making it chiral (tetrahedral) and optically active.

- Exception: Glycine is non-chiral because its R-group is just a Hydrogen atom (**-H**).
- Proteins ONLY contain L-amino acids.

6. Classification of Amino Acids (By Structure)

Based on the R-substituent, the 20 amino acids fall into 7 groups:

- Aliphatic: Hydrophobic & non-polar. Simple/branched chain.
  (Glycine, Alanine, Valine, Leucine, Isoleucine)
- Aromatic: Contain aromatic rings. They absorb UV light.
  (Phenylalanine, Tyrosine, Tryptophan)
- Acidic: Contain extra carboxyl groups.
  (Aspartic acid, Glutamic acid) and their amides (Asparagine, Glutamine).
- Basic: Basic in nature.
  (Lysine, Arginine, Histidine).
- Sulphur-containing: Have a sulphur atom.
  (Cysteine, Methionine).
- Hydroxyl-containing: Have an -OH group.
  (Serine, Threonine).

- *Imino Acid: Have an imino group (=**NH**) where the **α**-amino nitrogen is part of a ring. (Proline).*

7. Classification (By Nutritional Requirements)

Essential Amino Acids	*Non-Essential Amino Acids*
Cannot be synthesized by the body; MUST be obtained from diet. Required for proper growth.	*Synthesized in the body; do not strictly need to be consumed in the diet.*
List (10): Valine, Lysine, Arginine, Histidine, Isoleucine, Leucine, Methionine, Phenylalanine, Threonine, Tryptophan.	*List (10): Glycine, Alanine, Serine, Cysteine, Aspartic acid, Glutamic acid, Glutamine, Tyrosine, Proline, Asparagine.*

8. Physical Properties & Zwitterions

- *Solubility: All are water-soluble, but polarity affects exact solubility. Polar ones love water; non-polar ones dissolve better in organic solvents.*
- *Optical Activity: All (except Glycine) are optically active due to the chiral center.*
- *Acid-Base Behavior (Amphoteric): Contain both acidic (**-COOH**) and basic (**-NH_2**) groups, making them ampholytes.*

The Zwitterion (Dipolar Ion)

Amino acids rarely exist in neutral form. They usually exist as a hybrid molecule containing both positive and negative charges, resulting in a net charge of zero at the isoelectric point (pI).

pH 1-5 (Acidic) → *Cation (+)*

(Excess H^+ *:* ***-COOH*** *and* $-NH_3^+$*)*

At pI → *Zwitterion (Net Charge 0)*

($-COO^-$ *and* $-NH_3^+$*)*

pH 10-14 (Basic) → *Anion (-)*

(Excess OH^- *:* $-COO^-$ *and* $-NH_2$*)*